TPTG620 - Teaching Practice (Long Term)

Assignment Lesson Plan

Sr. No	Grade/Class	Subject	Topic
1	8th	Science	Structure of Nephron
2	9th	Physics	Pascal's Law
3	10th	Mathematics	Cube Roots of unity

LESSON PLAN # 01

1. **Subject:** Science
2. **Grade level:** 8th
3. **Text Book:** PTB
4. **Topic and Subtopic:**

Structure of Nephron

5.Time Duration: 40 mints

6.General Objective:

After acquiring knowledge about structure of nephron the students will be able to:

- Understand functional unit of kidney.
- Analyze different parts of nephron.
- Compare tubular and non tubular parts.

7.Specific Objective:

- Describe how nephron is functional unit of kidney.
- Compose and label the structure of nephron.
- Distinguish different parts of tubular structure of nephron.

8.Resources:

Different color markers, Table, pencil ,chart of label diagram of nephron.

9.Procedure:

(a).Introduction:

I will take test of internal structure of kidney and introduce about nephron .

(b). Teaching procedure:

i. I will use lecture method deliver topic of the day nephron.
ii. I will tell the students about the most important and functional part of kidney.
iii. I will explain the structure of nephron by drawing diagram on board.

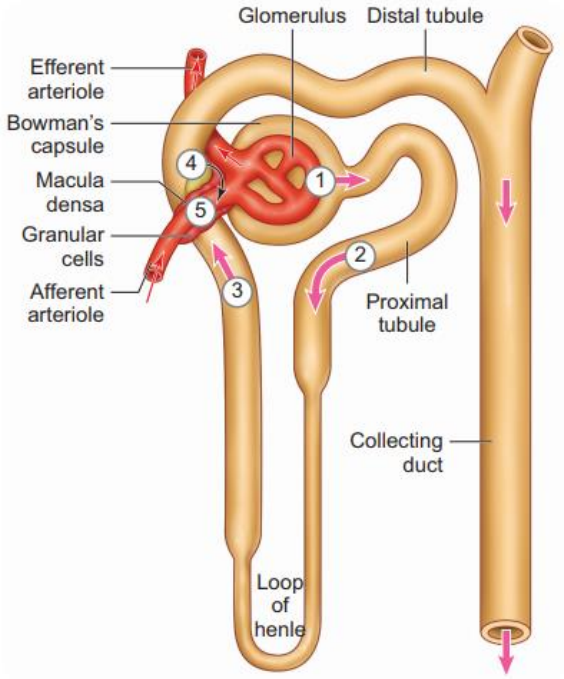


Figure 8.4 Structure of a Nephrons

iv. I will clearly discuss each part with definitions.

(c). Student's Participation:

I will ask students to share what they have learned from the structure of nephron. Also every student explain one part.

(d).Formative Check:

For formative check ,I will ask

- ✓ Which is the functional unit of kidney ?
- ✓ Differentiate between tubular and non tubular part?

(e). Summarizing and Closing:

I will briefly summarize the topic and conclude by clearing the confusion of students (If any).I will close the lecture by saying good luck for next class.

10.Assignment & Home Task:

The students can make assignment on following:

i. Draw the label diagram on registers.

ii. Write definition of each part.

11.Assessment of student's learning:

I will assess the concept of students by asking different questions like:

Q#1: What is meant by Loop of Henle?

Q#2: What is difference between proximal and distal convoluted tubule?

Q#3: Where urine is formed?

LESSON PLAN # 02

1.Subject: Physics
2.Grade level: 9th
3.Text Book: PTB
4.Topic and subtopic:
Topic: Pascal's Law
Subtopic: Applications of Pascal's Law

5.Time Duration: 40 mints

6.General Objective:

After taking the lecture students will be able to:

- Understand the effects of pressure.
- Explain and state Pascal's law.
- Understand the practical example of Pascal's law.
- Discuss about the working of hydraulic press.

7.Specific Objective:

- Describe Pascal's law and it's Applications.
- Discuss break system of vehicles.
- Perform an experiment to verify Pascal's law.

8.Resources:

Marker, Table, pencil, beaker having piston.

9.Procedure:

(a).Introduction:

Before starting lecture first I will give a brief review of previous topic **pressure in liquid and factors affecting it.**

I will tell students the reason of defining it again and connect them with today's topic that is Pascal's law.I will give a introduction of Pascal's Law and it's background.

(b).Teaching Method:

i. I must use lecture method for teaching so that students understand easily.
ii. I will first state Pascal's Law and tell students about its detail by giving examples.
iii. Then I will explain the practical applications of Pascal's Law that are Hydraulic press and brack system.
iv. To explain working of hydraulic press I will draw its diagram on board and relate its working with Pascal's Law.I will also tell them why it is called force multiplier.
v. Then I will elaborate that brack system of vehicles follows Pascal's law.

vi. I will describe the importance of Pascal's law in general and reason of this law .

(c).Students' Participation:

For involvement of students I will arrange a 5 minutes activity to clarify the Pascal's Law practically.

Take a cylinder having holes on each side. Fill it with water and press the piston.

I will show the students that water is coming out from holes with equal pressure and tell them that force applied on piston creates pressure on water which is then distributed to each side equally.

(d). Formative Check :

To check the understanding of students ,I will ask them during lecture

- State Pascals law?
- On which bodies Pascals law is applicable?

(e).Summarizing and Closing:

I will conclude the topic and give time to students to think about topic and discuss if there is any confusion.

10.Assignment & Home Task:

I will ask students to submit assignment on the following:

- Write down the practical applications of Pascal's law in daily life.
- Solve the sample problem 7.2 and exercise problem 7.7.

11.Assessment of student's learning:

To assess whether students have learned Pascal's Law completely I will ask different questions such as:

Q#1: What is the effect of height on pressure in liquid?

Q#2: Break systems work on the principle of Pascal's law. How can you justify it.

Q#3: Can you give any use of Pascal's law in daily life?

LESSON PLAN # 03

1. **Subject:** Mathematics
2. **Grade level: 10th**
3. **Text Book:** PTB
4. **Topic and Subtopic:**

   Topic: Cube Roots of unity

   Sub Topic: Recognize complex cube roots of unity , properties of cube roots of unity.

5.Time Duration: 40 mints

6.General Objective:

After learning about Cube Roots of Unity the students will be able to:

- Define Cube Roots of unity.
- Analyze different properties of cube roots of unity.
- Discuss about complex cube roots of unity.

7.Specific Objective:

- Recognize complex cube roots of unity.
- Justify that the product of three cube roots of unity is one.
- Describe the uses of properties of cube roots in problem solving

8.Resources:

Text Book, White board , marker , duster and pencil.

9.Procedure:

(a). Introduction:

To bridge up students concentration in the class first I will give review about quadratic equation and it's discriminant .Then I will introduce the topic of the day i.e cube roots and their properties.

(b). Teaching procedure:

i. I will use lecture method to explain the concept of Cube roots of unity.
ii. First I will ask from students about the roots of an equation then I will define the cube roots of unity and procedure to find it.
iii. I will tell the students that two out of three roots of unity are complex so these are called complex cube roots of unity.
iv. I will explain that each complex root us square of the other and will prove it on board.
v. I will also tell the students that addition of three roots of unity is always one .
vi. I will discuss another property of cube roots of unity that is each cube root is reciprocal of the other.
vii. At the end I will explain how can we use all these properties to solve different problems .

(c). Student's participation:

To involve students, I will make their groups and ask them to prove all properties on their registers and ask questions if there is any confusion.

(d).Formative Check:

For formative check I will ask from students how can they differ addition of cube roots of unity and multiplication of cube roots of unity? I will ask them to check what is the result if we add cube roots of unity.

(e) Summarizing and Closing:

I will summarize the whole lecture by explaining briefly about cube roots of unity and their properties and close the lecture by saying have a blessed day.

10.Assignment & Home Task:

I will ask students to submit assignment on following:

- Prove all the properties of cube roots of unity.
- Solve Exercise 2.2 with the help of these properties on your registers.

11.Assessment of student's learning:

I will assess the concept of students by asking different questions like:

Q#1: What do you know about complex cube roots of unity?

Q#2: Each complex cube root of unity is _____ of the other.

A. Square. B. Reciprocal. C. Inverse

Q#3: Some of all cube roots of unity is..?

A. 0. B. 1. C. both

Q#4: Define cube roots of unity.